

PROJECT DOCUMENTATION

REQUIREMENT ANALYSIS

EduGenie – Google Gemini Powered Learning Assistant

1. Functional Requirements

The system should allow students to ask academic questions, receive concise answers, simplify concepts, generate quizzes, summarize educational passages, and create structured learning paths.

2. Hardware Requirements

Computer or laptop with a modern processor; at least 4 GB RAM with 8 GB or more recommended; internet connection for Gemini API communication; keyboard, mouse, and display.

3. Software Requirements

Python 3.x, FastAPI, Uvicorn, Google Gemini / Google GenAI Python SDK, Jinja2, HTML5, CSS3, python-dotenv, and a modern web browser.

4. Backend Requirements

FastAPI application, template serving, POST form handling, Gemini communication, generated answer return, and CSS serving through /static.

5. Frontend Requirements

Project title and subtitle, question input area, Ask AI button, dynamic AI answer section, feature cards, and responsive layout.

6. Security Requirements

Keep the Gemini API key in .env, do not commit .env to version control, use .gitignore for secrets and temporary files, validate user input, avoid unnecessary personal information, and use HTTPS and server-side security controls in production.

7. Data Requirements

The current prototype does not require a traditional database for the basic question-answering flow. A future production version could store accounts, learning history, quiz scores, saved summaries, and progress.

8. Scope and Limitations

The project is an educational prototype and should not replace teachers, official textbooks, institutional guidance, or expert advice. AI responses can sometimes be inaccurate or incomplete.

Prepared for project documentation and academic demonstration.